Phy - III

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Sec-08

Ans. to the que. no. 1

Given,

$$\vec{F} = 3xy\hat{x} - 5z\hat{y} + 10x\hat{z}$$

Now,

$$\vec{F}.\vec{dr} = (3xy\hat{x} - 5z\hat{y} + 10x\hat{z})(dx\,\hat{x} + dy\,\hat{y} + dz\,\hat{z})$$

$$= 3xy\,dx - 5z\,dy + 10x\,dz$$

$$= 3(t^2+1)(2t^2)(2t)dt - 5(t^3)(4t)dt + 10(t^2+1)(3t^2)dt$$

$[\because x = t^2+1,\ y = 2t^2,\ z = t^3]$

$$= (12t^5 + 12t^3 - 20t^4 + 30t^4 + 30t^2)dt$$

$$= (12t^5 + 10t^4 + 12t^3 + 30t^2)dt$$

We know,

$$W = \int_C \vec{F}.dr$$

$$= \int_1^2 (12t^5 + 10t^4 + 12t^3 + 30t^2)dt$$

$$= \left[2t^6 + 2t^5 + 3t^4 + 10t^3\right]_1^2$$

$$= (128 + 64 + 48 + 80) - (2 + 2 + 3 + 10)$$

$$= 320 - 17$$

$$= 303 \text{ J}$$

Ans.

Ans. to the que. no. 2

Gravitational force between a particle of mass m_1 and one of mass m_2 is given by,

$$F(x) = \frac{k m_1 m_2}{x^2} \quad \text{--- (1)}$$

(a) → Potential energy $U(x)$

$$F = -\frac{dU}{dx}$$

$$U(x) = -\int F dx$$

$$= -\int \frac{k m_1 m_2}{x^2} dx$$

$$= -\frac{k m_1 m_2}{x} + C \qquad [x \to \infty,\ U(x) = 0]$$

$$\therefore C = 0$$

$$\therefore U(x) = -\frac{km_1m_2}{x} \quad \text{Ans.}$$

(b) → To increase the separation of the particles from $x = x_1$ to $x = x_1 + d$

$$W = \int_{x_1}^{x_1+d} F\,dx$$

$$= \int_{x_1}^{x_1+d} \frac{km_1m_2}{x^2}\,dx \qquad [\because \text{eqn } (1)]$$

$$= \frac{km_1m_2 d}{x_1(x_1+d)} \quad \text{Ans.}$$

Ans. to the que. no. 3

Given,

Small bucket, $m_1 = 4kg$

big bucket, $m_2 = 12kg$

Final velocity, $v_f = ?$

Initial velocity, $v_i = 0 ms^{-1}$

$h = 2m$

$h_0 = 0m$

We know,

$$\frac{1}{2} m_2 v_i^2 + m_2 g h_0 + \frac{1}{2} m_1 v_i^2 + m_1 g h_0 = \frac{1}{2} m_2 v_f^2 + m_2 g h + \frac{1}{2} m_1 v_f^2 + m_1 g h$$

$$0 + (12 \times 9.8 \times 2) + 0 + 0 = \frac{1}{2}(12) v_f^2 + 0 + \frac{1}{2}(4) v_f^2 + (4 \times 9.8 \times 2)$$

$$8 v_f^2 = 235.44 - 78.48$$

$$\therefore v_f = 4.4 ms^{-1}$$

$\therefore$ The speed with which this bucket strikes the floor is $4.4 ms^{-1}$ Ans.